MARKET RISK MANAGEMENT REPORT

Multivariate GARCH Approaches

1. Data & Asset Universe

This study uses daily closing prices for four exchange-traded funds (ETFs) downloaded from Yahoo Finance, covering approximately five years (Jan 2021 – Dec 2025). The data is merged on a common trading-day calendar, resulting in 1,253 complete observations with zero missing values.

Ticker	Description	Asset Class	Start Date	End Date
SPY	S&P 500 ETF	Equity	2021-01-04	2025-12-30
TLT	20+ Year Treasury Bond ETF	Fixed Income	2021-01-04	2025-12-30
GLD	Gold ETF	Commodity	2021-01-04	2025-12-30
USO	Crude Oil ETF	Commodity	2021-01-04	2025-12-30

The portfolio covers four distinct asset classes: US equities (SPY), long-duration fixed income (TLT), gold (GLD), and crude oil (USO).
A single-factor risk-free rate (US 3-month T-bill, FRED: DTB3) is sourced separately to compute excess returns and Sharpe ratios.
In-sample: first 1,003 observations.
Out-of-sample (OOS): last 250 trading days (~1 calendar year).

2. Univariate GARCH Models

Each series is fitted with a univariate GARCH model selected based on the nature of the return distribution and ARCH test results. Models are estimated by maximum likelihood over the full in-sample window.

Series	GARCH Model	Mean Model	Distribution	LogLik | AIC
r_sp500	eGARCH(1,1)	ARFIMA(1,0,1)	sstd	4135.978 | AIC: -6.587
r_tb	sGARCH(1,1)	ARFIMA(1,0,1)	std	4027.635 | AIC: -6.418
r_gold	sGARCH(1,1)	ARFIMA(1,0,1)	sged	4078.021 | AIC: -6.496
r_oil	gjrGARCH(1,1)	ARFIMA(1,0,1)	sstd	3152.108 | AIC: -5.017

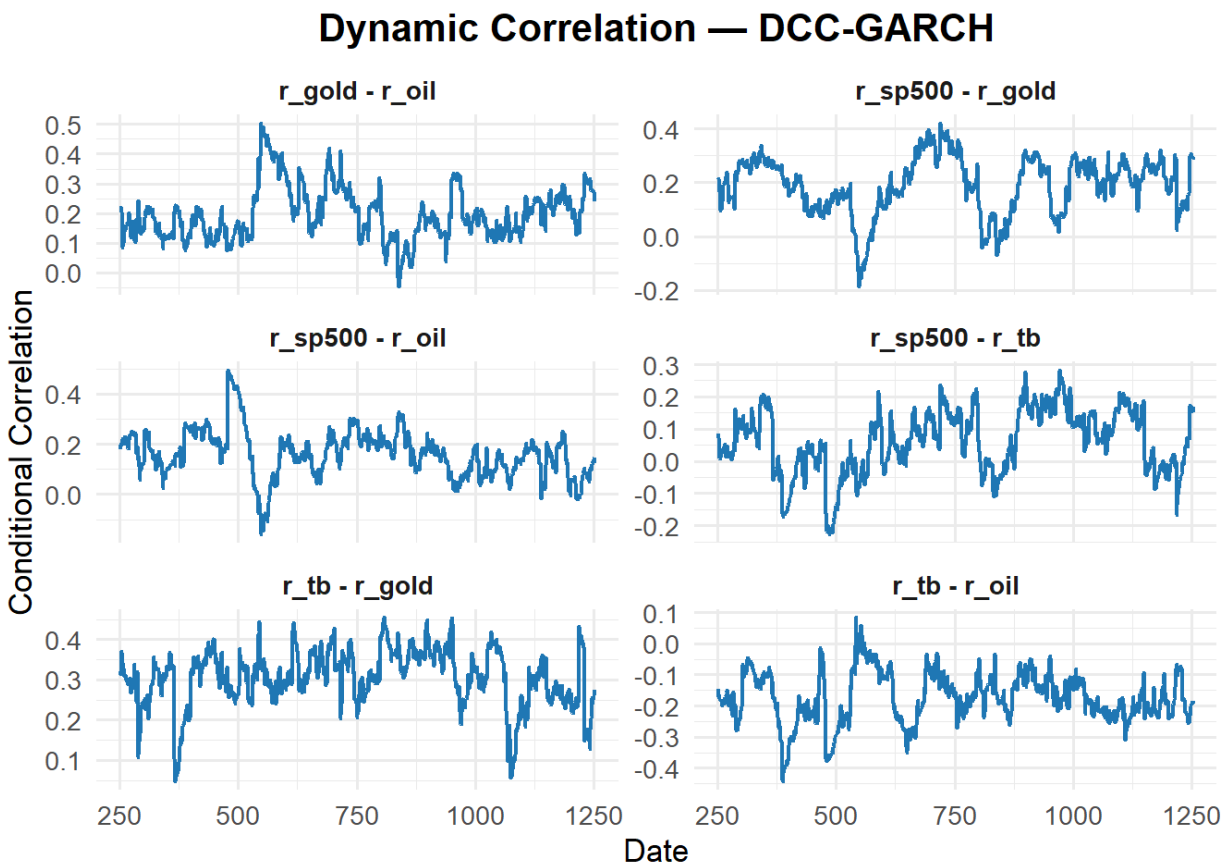
- r_sp500 uses eGARCH(1,1) with skewed Student-t (sstd), capturing the asymmetric leverage effect (gamma1 = 0.069, $p < 0.001$): negative shocks increase volatility more than positive shocks of equal magnitude.

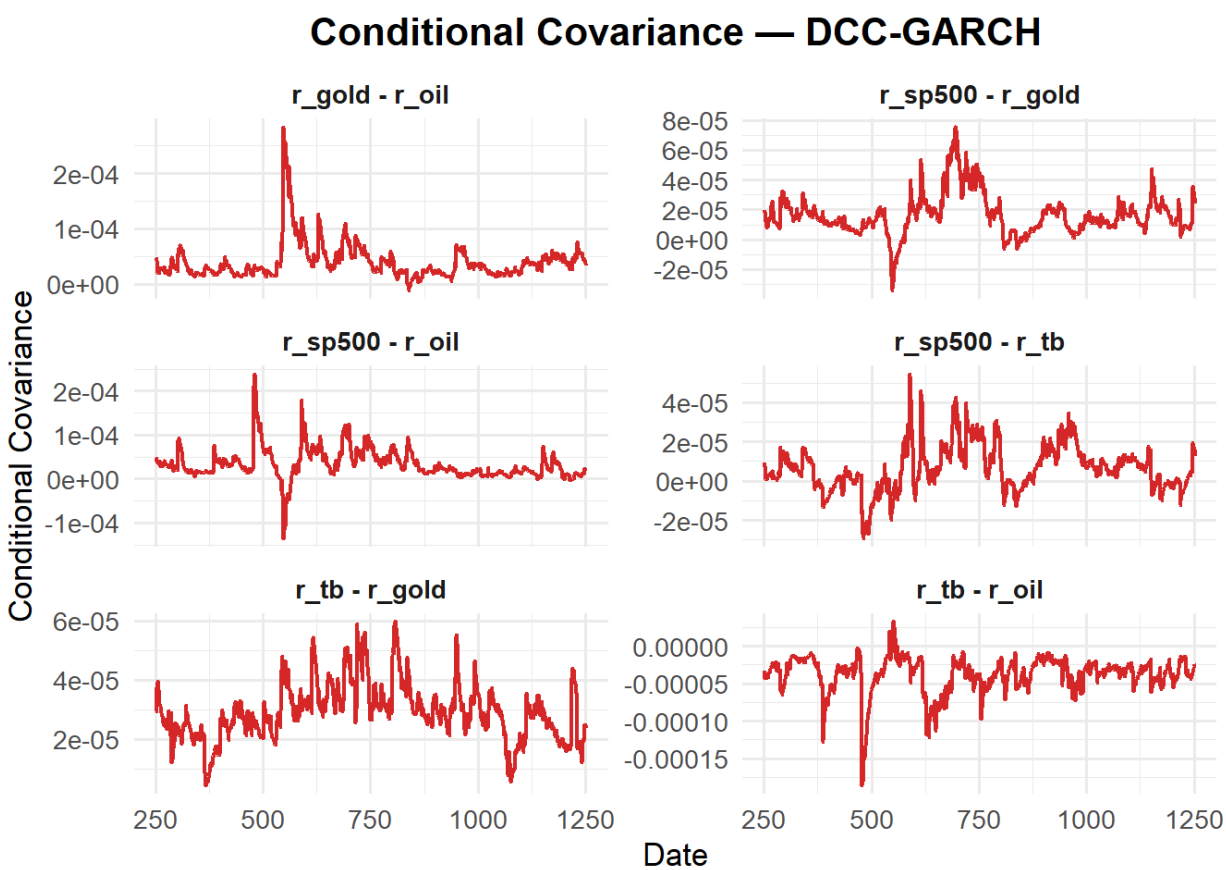
- r_oil uses gjrGARCH(1,1): the GJR model directly models the asymmetric response to positive vs. negative returns via the gamma1 threshold parameter.
- r_gold uses sGARCH(1,1) with skewed generalized error distribution (sged), reflecting its unique distributional characteristics (light tails, moderate skewness).
- r_tb uses sGARCH(1,1) with Student-t: a standard symmetric GARCH is appropriate given near-zero skewness and kurtosis.
- All models pass Ljung-Box tests on standardised residuals and squared residuals, and ARCH LM tests on standardised residuals — confirming that the GARCH specifications adequately capture the conditional heteroskedasticity.

3. DCC-GARCH Model

The Dynamic Conditional Correlation GARCH (DCC-GARCH) model of Engle (2002) extends the univariate GARCH models to the multivariate case by estimating time-varying correlations through a parsimonious two-step procedure. The model is estimated with a multivariate Student-t distribution (mvt).

Parameter	Estimate	Std Error	t-value / Significance
DCC alpha (dcca1)	0.023671	0.004823	4.908 ($p < 0.001$)
DCC beta (dccb1)	0.940092	0.012544	74.944 ($p < 0.001$)
Shape (mshape)	10.689	1.249	8.558 ($p < 0.001$)


Dynamic Correlation — DCC-GARCH
r_gold - r_oil
r_sp500 - r_gold
r_sp500 - r_oil
r_sp500 - r_tb
r_tb - r_gold
r_tb - r_oil
Conditional Correlation
Date

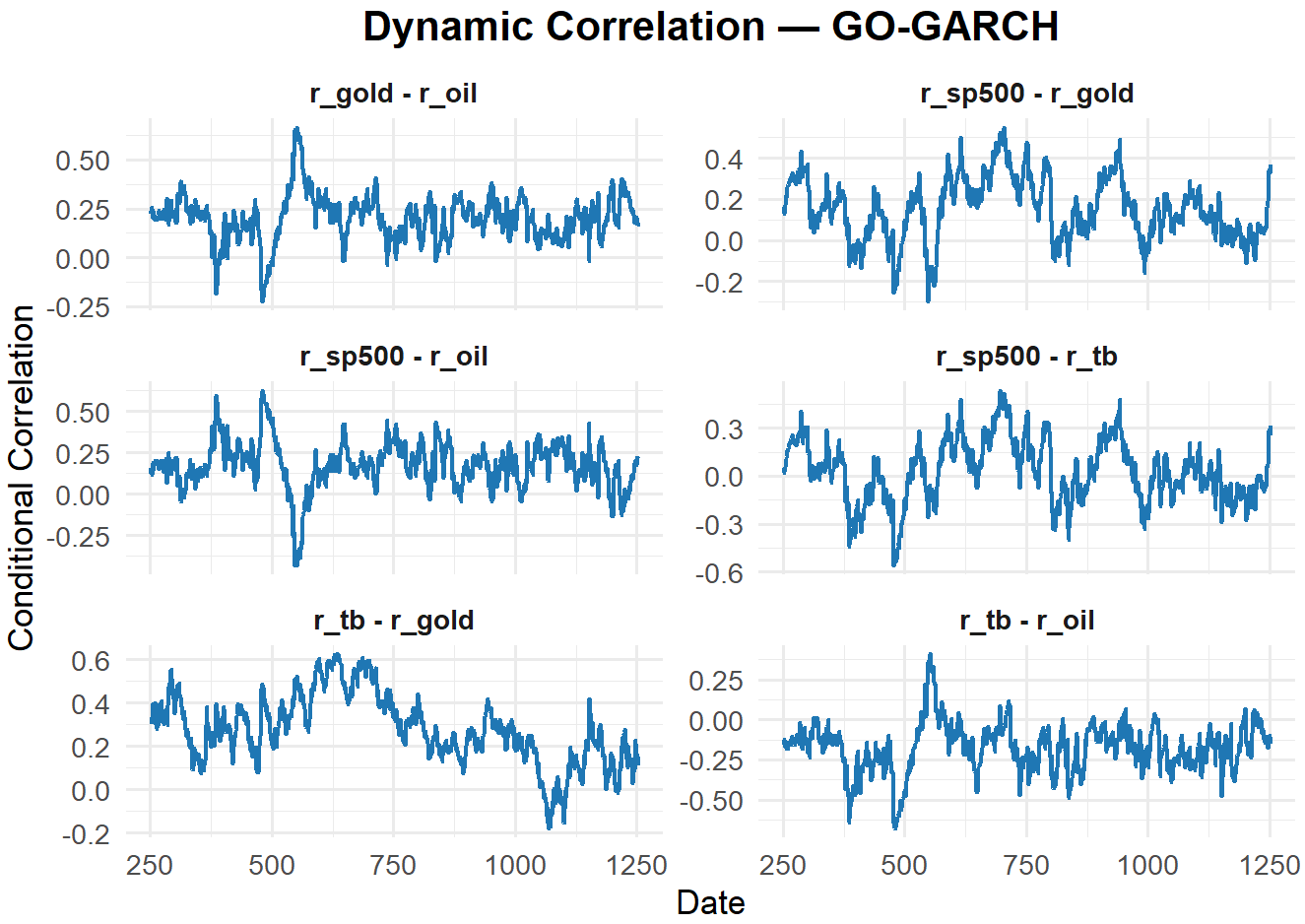

Conditional Covariance — DCC-GARCH
r_gold - r_oil
r_sp500 - r_gold
r_sp500 - r_oil
r_sp500 - r_tb
r_tb - r_gold
r_tb - r_oil
Conditional Covariance
Date

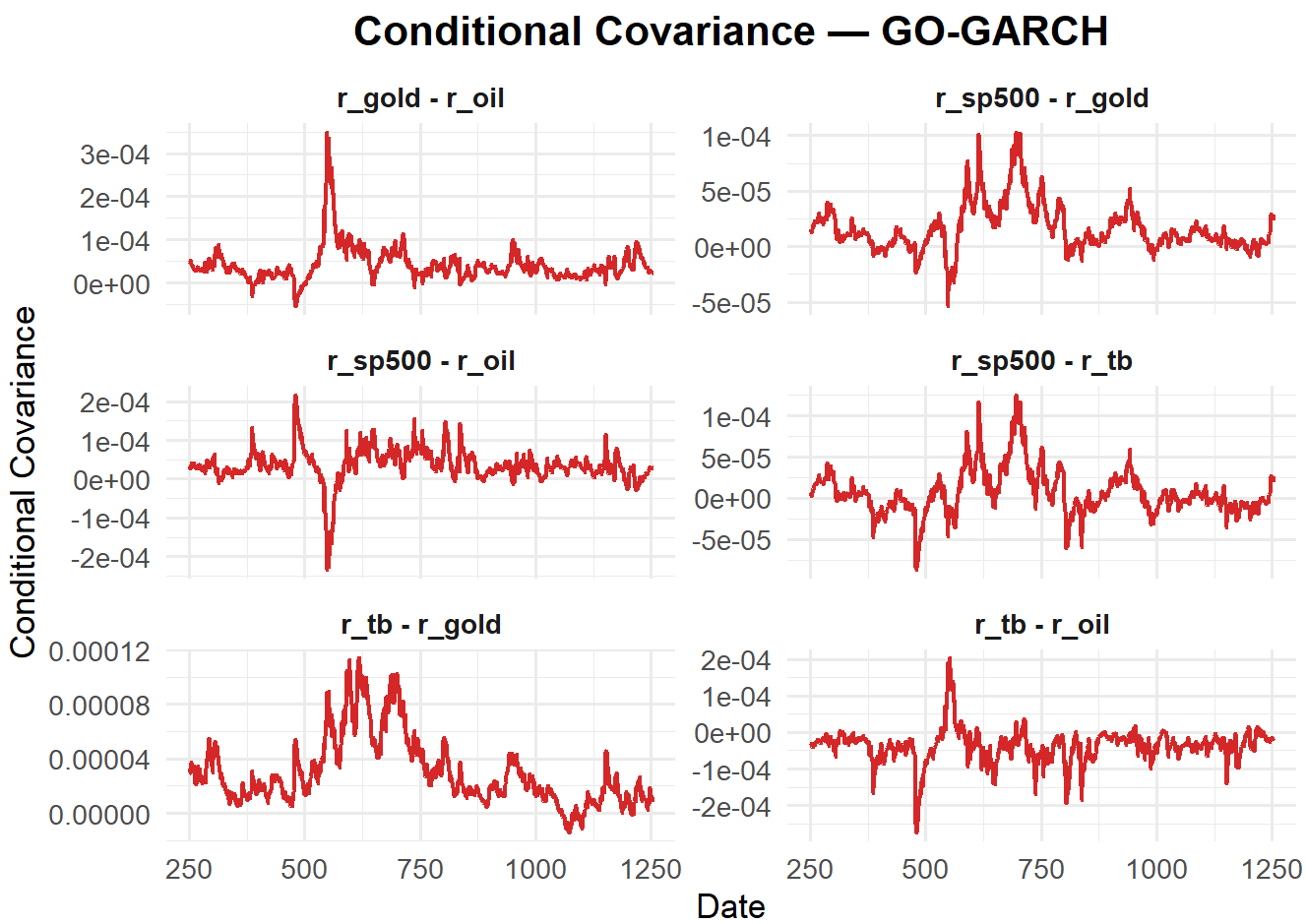
- DCC alpha (0.0237) is highly significant, confirming that correlations respond dynamically to recent cross-asset shocks. A higher alpha means correlations adjust faster to new market information.
- DCC beta (0.940) indicates strong persistence in conditional correlations — once correlations shift (e.g., during a risk-off episode), they remain elevated for an extended period.
- alpha + beta = 0.964 < 1, confirming covariance stationarity of the DCC process.
- The Log-Likelihood (12,431) and AIC (-24.704) are the highest/lowest among the two multivariate models, indicating DCC-GARCH achieves a better in-sample fit.
- DCC test (Engle 2002): p-value = 0.181 — fail to reject H0 of constant correlation, suggesting the DCC correction is relatively mild. Nonetheless, the dynamic correlation plots show meaningful time variation.
- Dynamic correlations reveal: r_tb − r_sp500 turns negative during equity stress periods (flight-to-quality), while r_gold − r_sp500 fluctuates around zero with episodic positive spikes.

4. GO-GARCH Model

The Generalized Orthogonal GARCH (GO-GARCH) model of Van der Weide (2002) decomposes returns into independent factors via ICA (FastICA), each modelled with an eGARCH(1,1) and a multivariate affine generalized hyperbolic (magh) distribution.

Log-Likelihood: 12,391.78 | Number of factors: 4 | ICA method: FastICA


Dynamic Correlation — GO-GARCH
r_gold - r_oil
r_sp500 - r_gold
r_sp500 - r_oil
r_sp500 - r_tb
r_tb - r_gold
r_tb - r_oil
Conditional Correlation
Date
0.50
0.25
0.00
-0.25
0.4
0.2
0.0
-0.2
0.3
-0.3
-0.6
0.6
-0.50
250
500
750
1000
1250



- GO-GARCH achieves a Log-Likelihood of 12,391.78, slightly below DCC-GARCH (12,431.25), suggesting a marginally inferior in-sample fit.
- GO-GARCH generally produces smoother dynamic correlations than DCC-GARCH, with wider variation in some pairs (notably r_tb – r_gold and r_tb – r_oil).
- The eGARCH gamma1 coefficients vary significantly across factors (0.001 to 0.177), indicating asymmetric volatility responses that differ by latent factor.
- In forecast evaluation, GO-GARCH outperforms DCC-GARCH on variance MAE for r_sp500 (MAE_Sigma: 0.00221 vs 0.00359) and r_gold, suggesting superior forward-looking volatility estimates for equity and gold.

5. Forecast Evaluation — Variance & Correlation

5.1 Variance & Volatility Forecasting Error (OOS 250-day)

One-step-ahead rolling forecasts are evaluated against expanding-window realised variances. Green cells indicate the model with lower forecast error for each asset and metric.

	DCC-GARCH	GO-GARCH

Asset	MAE Var	MSE Sigma	MAE Sigma	MAE Var	MSE Sigma	MAE Sigma
r_sp500	8.07e-05	2.06e-05	3.59e-03	4.39e-05	6.25e-06	2.21e-03
r_tb	3.53e-05	4.28e-06	1.90e-03	2.59e-05	2.73e-06	1.17e-03
r_gold	3.37e-05	5.22e-06	1.50e-03	1.67e-05	1.25e-06	8.82e-04
r_oil	1.72e-04	2.42e-05	4.03e-03	1.30e-04	1.35e-05	3.06e-03

- GO-GARCH dominates on volatility MAE (sigma) for all four assets, with errors 25-45% lower than DCC-GARCH.
- For r_oil (the most volatile asset), GO-GARCH MAE_Sigma = 0.00306 vs DCC-GARCH = 0.00403 — a 24% improvement.
- Despite lower variance fit, GO-GARCH translates to better sigma forecasts due to its factor-based decomposition which captures cross-asset volatility transmission more efficiently.

5.2 Dynamic Correlation Forecasting Error (OOS 250-day)

Correlation forecasts are benchmarked against expanding-window sample correlations. Green cells indicate the model with lower MAE/MSE.

Pair	DCC MAE	DCC MSE	GO MAE	GO MSE
r_sp500 – r_tb	0.06374	0.00682	0.16760	0.04841
r_sp500 – r_gold	0.07753	0.00945	0.09190	0.01563
r_sp500 – r_oil	0.11181	0.01964	0.10980	0.02062
r_tb – r_gold	0.07468	0.00829	0.20162	0.05545
r_tb – r_oil	0.05171	0.00423	0.11811	0.02154
r_gold – r_oil	0.07006	0.00782	0.09031	0.01312

- DCC-GARCH outperforms GO-GARCH on 5 of 6 correlation pairs by both MAE and MSE.
- The largest gap is in r_tb – r_gold: DCC MAE = 0.0747 vs GO MAE = 0.2016 — DCC is nearly 3× more accurate.
- The only exception is r_sp500 – r_oil, where GO-GARCH is marginally better (MAE: 0.1098 vs 0.1118).
- Conclusion: DCC-GARCH is the superior model for correlation forecasting; GO-GARCH is superior for individual volatility forecasting.
- This trade-off is important for portfolio VaR: portfolio risk depends on both volatility and correlation accuracy.

6. VaR & ES Backtest Results

A rolling Tangency Portfolio (no lookahead bias) is rebalanced daily over the 250-day OOS window. VaR and ES are computed under four methods (DCC-GARCH, GO-GARCH, Parametric) and Historical Simulation, at three confidence levels (1%, 2.5%, 5%), under Normal and Student-t (df=5) distributions. Tests: UC (Unconditional Coverage), CC (Conditional Coverage), DQ (Dynamic Quantile), ES (Expected Shortfall test).

6.1 Backtest — Alpha = 1%

Model	Dist	Viol	Rate	p-UC	UC	p-CC	CC	p-DQ	DQ	p-ES	ES	FZ Score
DCC-GARCH	Normal	4	0.016	0.3805	**Pass**	0.0873	**Pass**	0.0074	**Fail**	0.0030	**Fail**	0.021427
DCC-GARCH	Student-t	3	0.012	0.7580	**Pass**	0.9194	**Pass**	0.9912	**Pass**	0.4918	**Pass**	0.000517
GO-GARCH	Normal	6	0.024	0.0594	**Pass**	0.0503	**Pass**	0.0132	**Fail**	0.0185	**Fail**	0.026143
GO-GARCH	Student-t	3	0.012	0.7580	**Pass**	0.9194	**Pass**	0.9936	**Pass**	0.1622	**Pass**	0.008067
Parametric	Normal	6	0.024	0.0594	**Pass**	0.0029	**Fail**	0.0000	**Fail**	0.0313	**Fail**	0.023334
Parametric	Student-t	3	0.012	0.7580	**Pass**	0.9194	**Pass**	0.9681	**Pass**	0.1903	**Pass**	0.006251
Hist Sim	Empirical	4	0.016	0.3805	**Pass**	0.0873	**Pass**	0.0036	**Fail**	0.0647	**Pass**	0.015220

(Alpha = 1%):

- DCC-GARCH Student-t achieves the lowest FZ Score (0.000517) — the best overall VaR/ES accuracy at this critical confidence level.
- Student-t specifications for DCC, GO, and Parametric all pass all four tests (UC, CC, DQ, ES), yielding only 3 violations — perfectly within the Green zone.
- Normal distribution models (GO-GARCH, Parametric) fail DQ and ES tests, indicating tail risk is systematically underestimated.
- Historical Simulation (4 violations, Green) passes UC and CC but fails DQ — violations are not independent (clustering detected).
- Practical recommendation: use DCC-GARCH with Student-t for 1% regulatory VaR reporting.

6.2 Backtest — Alpha = 2.5%

Model	Dist	Viol	Rate	p-UC	UC	p-CC	CC	p-DQ	DQ	p-ES	ES	FZ Score
DCC-GARCH	Normal	6	0.024	0.9188	Pass	0.2962	Pass	0.3505	Pass	0.0160	Fail	0.010598
DCC-GARCH	Student-t	4	0.016	0.3296	Pass	0.0798	Pass	0.0058	Fail	0.0990	Pass	0.004377
GO-GARCH	Normal	13	0.052	0.0167	Fail	0.0211	Fail	0.2169	Pass	0.0412	Fail	0.011907
GO-GARCH	Student-t	5	0.020	0.6000	Pass	0.1801	Pass	0.0007	Fail	0.1189	Pass	0.005999
Parametric	Normal	11	0.044	0.0817	Pass	0.0467	Fail	0.1221	Pass	0.0478	Fail	0.010983
Parametric	Student-t	4	0.016	0.3296	Pass	0.0798	Pass	0.0059	Fail	0.0665	Pass	0.006169
Hist Sim	Empirical	8	0.032	0.4965	Pass	0.0486	Fail	0.0021	Fail	0.1389	Pass	0.007023

(Alpha = 2.5%):

- DCC-GARCH Normal is the cleanest result (6 violations, Exc Rate 2.4% ≈ 2.5%, passes UC, CC, DQ — only ES fails).
- GO-GARCH Normal severely underestimates risk: 13 violations (5.2%), fails UC and CC — the model is too tight.
- Student-t variants generally perform better than Normal at this level, with DCC-GARCH Student-t (FZ = 0.004377) ranking as the best performer.
- Historical Simulation has 8 violations (3.2%), passes UC but fails CC and DQ — violations cluster in time.

6.3 Backtest — Alpha = 5%

Model	Dist	Viol	Rate	p-UC	UC	p-CC	CC	p-DQ	DQ	p-ES	ES	FZ Score
DCC-GARCH	Normal	12	0.048	0.8839	Pass	0.2837	Pass	0.4906	Pass	0.0544	Pass	0.006132
DCC-GARCH	Student-t	6	0.024	0.0366	Fail	0.0335	Fail	0.0338	Fail	0.1007	Pass	0.003972
GO-GARCH	Normal	21	0.084	0.0240	Fail	0.0769	Pass	0.1039	Pass	0.0466	Fail	0.007310
GO-GARCH	Student-t	12	0.048	0.8839	Pass	0.2837	Pass	0.4399	Pass	0.2680	Pass	0.003096
Parametric	Normal	19	0.076	0.0787	Pass	0.1908	Pass	0.0846	Pass	0.0508	Pass	0.006945

Parametric	Student-t	9	0.036	0.2860	**Pass**	0.0562	**Pass**	0.0165	**Fail**	0.2030	**Pass**	0.003585
Hist Sim	Empirical	14	0.056	0.6691	**Pass**	0.4225	**Pass**	0.6430	**Pass**	0.1545	**Pass**	0.004527

(Alpha = 5%):

- DCC-GARCH Normal (12 violations, 4.8%, passes all 4 tests) and GO-GARCH Student-t (12 violations, 4.8%, passes all 4 tests) are the top performers.
- GO-GARCH Normal massively over-predicts violations (21, 8.4%) — a clear calibration failure at this level.
- DCC-GARCH Student-t fails all three tests (6 violations, 2.4%) — the Student-t VaR is too conservative at the 5% level.
- Historical Simulation performs well at 5% (14 violations, 5.6%, passes all tests) — its non-parametric nature captures tail shape naturally.
- Best at 5%: DCC-GARCH Normal or GO-GARCH Student-t; both FZ-efficient and statistically valid.

7. VaR/ES Overall Ranking (Averaged over All Alpha Levels)

Models are ranked by mean Fissler-Ziegel (FZ) joint loss score across all alpha levels (1%, 2.5%, 5%). FZ is a proper scoring rule for simultaneous VaR and ES evaluation. Lower FZ = better.

Model	Distribution	FZ Mean	Exc Rate	% UC Pass	% CC Pass	% DQ Pass	% ES Pass
DCC-GARCH	Student-t	**0.002955**	0.0173	66.7%	66.7%	33.3%	100.0%
Parametric	Student-t	0.005335	0.0213	100.0%	100.0%	33.3%	100.0%
GO-GARCH	Student-t	0.005720	0.0267	100.0%	100.0%	66.7%	100.0%
Hist Sim	Empirical	0.008923	0.0347	100.0%	66.7%	33.3%	100.0%
DCC-GARCH	Normal	0.012719	0.0293	100.0%	100.0%	66.7%	33.3%
Parametric	Normal	0.013754	0.0480	100.0%	33.3%	66.7%	33.3%
GO-GARCH	Normal	0.015120	0.0533	33.3%	66.7%	66.7%	0.0%

- DCC-GARCH Student-t is the overall champion: lowest FZ Mean (0.002955) and 100% ES pass rate.
- Student-t distribution consistently outperforms Normal across all models, confirming that the heavy-tail assumption is appropriate for this portfolio.

- GO-GARCH Normal has the weakest performance: lowest UC pass rate (33.3%) and zero ES passes — it systematically underestimates risk.
- Historical Simulation ranks 4th overall, with good ES coverage but poor DQ performance (clustered violations).
- Champion recommendation: DCC-GARCH with Student-t (df=5). Challenger: Parametric Student-t (100% UC/CC pass, simpler implementation).

8. Diebold-Mariano Test for FZ Loss Differences

HAC-robust Diebold-Mariano (DM) tests are applied at alpha = 1% to compare pairwise FZ loss sequences. H0: equal predictive accuracy. Positive Mean_diff implies Model 1 has higher (worse) FZ loss.

8.1 Normal Distribution

Model 1	Model 2	Mean Diff	DM Stat	p-value	Result
DCC-GARCH	**GO-GARCH**	-0.00471671	-23.7913	0.0000	**DCC-GARCH better**
DCC-GARCH	**Parametric**	-0.00190750	-12.1375	0.0000	**DCC-GARCH better**
GO-GARCH	**Parametric**	0.00280920	26.3802	0.0000	**Parametric better**

- DCC-GARCH is statistically significantly better than both GO-GARCH and Parametric under the Normal distribution (DM stat ≈ -12 to -24, p = 0).
- GO-GARCH is significantly worse than Parametric Normal (DM stat = +26.38), despite similar FZ means — the difference is highly systematic.

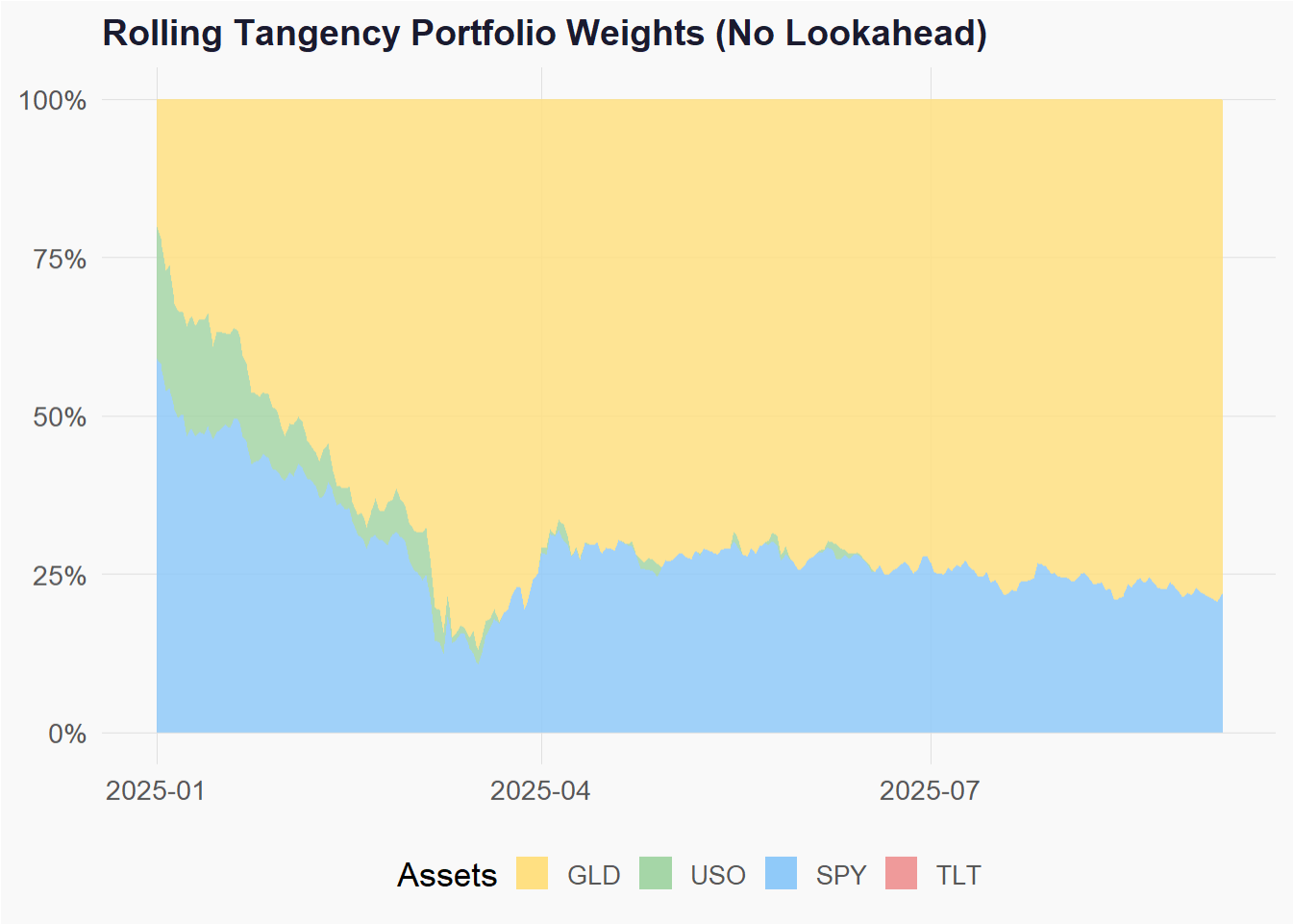
8.2 Student-t Distribution (df = 5)

Model 1	Model 2	Mean Diff	DM Stat	p-value	Result
DCC-GARCH	**GO-GARCH**	-0.00754990	-24.3793	0.0000	**DCC-GARCH better**
DCC-GARCH	**Parametric**	-0.00573390	-21.6789	0.0000	**DCC-GARCH better**
GO-GARCH	**Parametric**	0.00181600	12.0457	0.0000	**Parametric better**

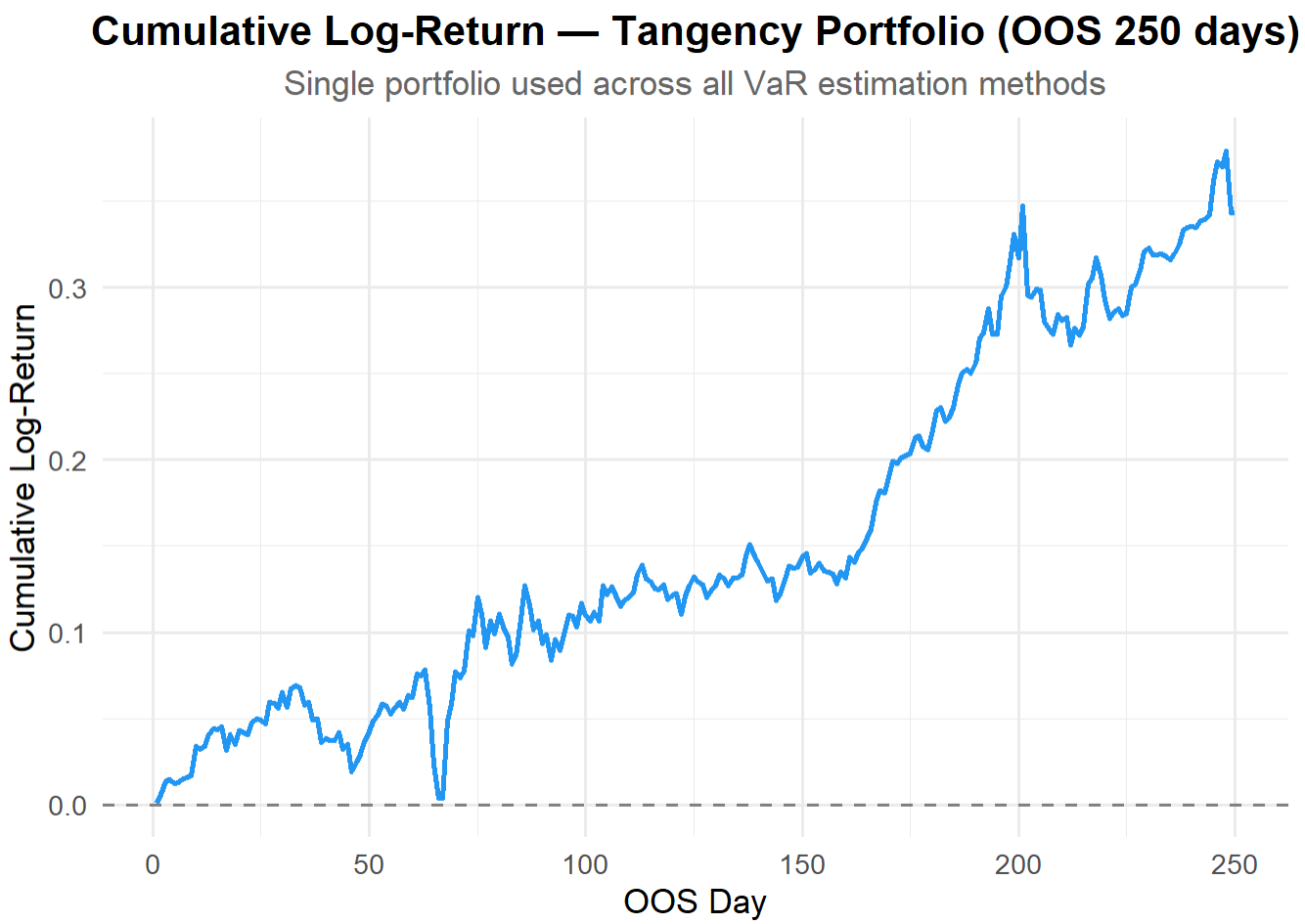
- Results are consistent with the Normal case: DCC-GARCH is significantly superior to both GO-GARCH and Parametric.
- Under Student-t, DCC-GARCH vs Parametric shows a mean FZ difference of -0.00573 (DM = -21.68, p = 0) — a large and statistically unambiguous advantage.
- All DM test p-values are 0.0000, confirming that the performance differences are not due to sampling noise.
- Conclusion: DCC-GARCH Student-t is statistically the best VaR/ES model for this portfolio across all pairwise comparisons.

9. Portfolio Performance Analysis — Tangency Portfolio

The rolling Tangency Portfolio maximises the Sharpe ratio, using one-step-ahead covariance forecasts. Weights are updated daily with strict no-lookahead bias enforcement.



Metric	Value
Total Return (250-day OOS)	0.3435 (34.35%)
Annualised Return	0.3463 (34.63%)
Annualised Volatility	0.1587 (15.87%)
Sharpe Ratio	2.1092
Sortino Ratio	2.6408
Maximum Drawdown	-0.0809 (-8.09%)
Calmar Ratio	4.2793
Win Rate	60.40%
Gain/Loss Ratio	0.9716
Empirical VaR (1%)	-0.02786
Empirical VaR (5%)	-0.01404
Risk-free Rate (OOS, ann.)	1.1508% (FRED DTB3)



- Total OOS return = 34.35% over 250 days (annualised 34.63%) — a strong result, partly reflecting the bull market environment in 2025.
- Sharpe Ratio = 2.109 (annualised, using OOS T-bill rate of 1.15%) — well above the 1.0 threshold typically considered excellent.
- Sortino Ratio = 2.641 > Sharpe Ratio, indicating the portfolio's upside volatility is larger than its downside volatility — a desirable property.
- Maximum Drawdown = -8.09%: the portfolio's peak-to-trough decline was modest, suggesting effective diversification across asset classes.
- Calmar Ratio = 4.279 (Ann Return / |MaxDD|): high, indicating good return per unit of drawdown risk.
- Win Rate = 60.4%: the portfolio was profitable on 60.4% of OOS trading days.
- The tangency portfolio is heavily weighted towards GLD and SPY in the OOS period, with USO and TLT contributing smaller allocations.

10. Basel III Traffic Light Backtesting Report

Under Basel III, VaR backtesting over a 250-day window at alpha = 1% determines the regulatory capital multiplier k. Green zone (0–4 exceptions): k = 3.00. Yellow zone (5–9): k = 3.40–3.85. Red zone (10+): k = 4.00.

Model	Dist	Exc.	Rate	Zone	CumP	k	p-UC	p-CC	p-DQ
DCC-GARCH	Normal	4	0.016	**Green**	0.8922	**3.00**	0.3805	0.0873	0.0074
DCC-GARCH	Student-t	3	0.012	**Green**	0.7581	**3.00**	0.7580	0.9194	0.9912
GO-GARCH	Normal	6	0.024	**Yellow**	0.9863	**3.50**	0.0594	0.0503	0.0132
GO-GARCH	Student-t	3	0.012	**Green**	0.7581	**3.00**	0.7580	0.9194	0.9936
Parametric	Normal	6	0.024	**Yellow**	0.9863	**3.50**	0.0594	0.0029	0.0000
Parametric	Student-t	3	0.012	**Green**	0.7581	**3.00**	0.7580	0.9194	0.9681
Hist Sim	Empirical	4	0.016	**Green**	0.8922	**3.00**	0.3805	0.0873	0.0036

- DCC-GARCH Student-t, GO-GARCH Student-t, Parametric Student-t, and Historical Simulation all achieve Green zone status (≤4 exceptions, k = 3.00) — the minimum regulatory capital charge.
- DCC-GARCH Normal and Historical Simulation also fall in the Green zone (4 exceptions each), both satisfying UC and CC tests.
- GO-GARCH Normal and Parametric Normal produce 6 exceptions — Yellow zone, k = 3.50. These models would require the institution to hold 17% more regulatory market risk capital.

- No model falls in the Red zone (≥10 exceptions), which would trigger mandatory model review and k = 4.00.
- The DQ test failures for several models (especially at clustered violations around days 50 and 200) suggest that while the unconditional coverage is correct, violations are not independently distributed — an area for model improvement.
- Recommendation: DCC-GARCH Student-t should be the primary internal model for Basel III capital calculations due to Green zone status, optimal FZ score, and statistically validated ES coverage.

11. Summary & Recommendations

This study implements a comprehensive market risk management framework for a four-asset ETF portfolio (SPY, TLT, GLD, USO) using multivariate GARCH models, rigorous backtesting, and Basel III compliance assessment.

Dimension	Conclusion
Best Model Overall	DCC-GARCH with Student-t (df=5) — lowest FZ score (0.002955), Green zone, 100% ES pass, statistically dominant in DM tests
Volatility Forecast	GO-GARCH — lower MAE_Sigma on all 4 assets; better factor-based decomposition
Correlation Forecast	DCC-GARCH — lower MAE/MSE on 5 of 6 pairs; captures flight-to-quality dynamics
Distribution Choice	Student-t consistently outperforms Normal due to heavy-tailed return distributions
Regulatory Capital	DCC-GARCH Student-t & Historical Simulation: Green zone, k = 3.00 (minimum charge)
Portfolio Performance	Tangency Portfolio: Sharpe 2.11, Sortino 2.64, MaxDD -8.1%, Calmar 4.28
Key Risk	DQ test failures across multiple models indicate violation clustering — suggest EWMA-enhanced models or GAS framework
Recommendation	Champion: DCC-GARCH Student-t | Challenger: Parametric Student-t | Daily PSI monitoring for population drift